## PHYSICS

1. B) $[MT^{-1}]$ | Explanation: In the given exponential term the exponent $\frac{\beta t}{m}$ must be entirely dimensionless. Therefore, the dimensions of $\beta$ must satisfy $[\beta]$ = [M]/[T], therefore $\beta = [MT^{-1}]$

2. A) $\alpha > \theta_0$ | Explanation: $\tau = I\alpha = 5 \times 10^{-3} \times 2\pi \times \frac{20}{10} = 2\pi \times 10^{-2} Nm$

3. A) a = gtan$\theta$ | Explantion: $tan\theta = \frac{ma}{mg} => a = gtan\theta$

4. D) d/2 | Explanation: d = $u^2/4$, $H_{max} = u^2/2g$ = d/2

5. A) 16ΔP | Explanation: According to Poiseuile's law, the fluid resistance $R_f$ of a pipe is given by $R_f = \frac{8\eta L}{\pi r^4}$ . Because the tubes are in series, the volumetric flow rate Q is identical. The resitance ratio $R_A/R_A = (r_B / r_A)^4 = (2/1)^4 = 16$, $P_A = 16\Delta P$

6. B) ¼ $m\omega^2A^2$ | Explanation: The kinetic energy at any instant t is K(t) = ½ $mv^2 = \frac{1}{2}m\omega^2A^2\cos^2(\omega t + \phi)$. To find the time average over one period T, we integrate K(t) and divide by T. The average value of a squared sinusoidal function $\cos^2(\theta)$ over one full cycle is strictly 1/2. Thus, $K_{avg}$ = 1/2 $m\omega^2A^2 * 1/2$ = ¼ $m\omega^2A^2$

7. A) $4GM/9R^2$ | Explanation: A point located at a distance of 1.5R is strictly outside the inner sphere of radius R but inside the outer spherical shell of radius 2R. According to Newton's Shell The orem, a uniform spherical shell exerts no net gravitational force on any object located inside it. Therefore, the entire 3M mass contributes nothing. The field is completely produced by the inner mass M. $g_{net} = GM/r^2 = GM/(1.5R)^2 = 4GM/9R^2$.

8. C) 0.5 | Explanation: Poisson's ratio σ correlates lateral strain to longitudinal strain. The frac tional change in volume for an elastic material subjected to uniaxial stress is given by $\Delta V/V = \epsilon_l(1 - 2\sigma)$, where $\epsilon_l$ is the longitudinal strain. A material that is perfectly in compressible maintains a constant volume under stress, meaning ΔV = 0. Therefore, $(1 - 2\sigma) = 0 \Rightarrow \sigma = 0.5$.

9. B) $2\pi T^2/\rho g$ | Explanation: By capillary action, the height the liquid rises is h = $2T/r\rho g$. The total mass of the elevated liquid column is m = volume × density = $(\pi r^2 h)\rho$. The center of mass of this uniform column is located at a height h/2. The gravitational potential energy is U = mgh/2 =$(\pi r^2 h\rho)gh^2 = 1/2\pi r^2\rho gh^2$. Substituting h into the equation: U = $1/2\pi r^2\rho g$ $(4T^2$ $r^2\rho^2g^2) = 2\pi T^2\rho g$ .

10. C) 3g | Explanation: For a bob to just successfully complete a vertical circle (tension at apex is zero), the velocity at the lowest point must be $v_L = \sqrt{5gL}$. Using conservation of mechanical energy from the lowest point to the horizontal position (where height gained is L): ½ $mv_L^2$= $1/2mv_H^2$ + mgL. Substituting $v_L^2$ = 5gL yields 5/2 mgL = $1/2mv_H^2$ + mgL ⇒ $v_H^2$ = 3gL. At the horizontal starting position, all centripetal acceleration is supplied directly by tention. It's magnitude is $a_c = v_H^2/L$ = 3gL/L = 3g

11. C) 4 : 1 | Explanation: By Stefan–Boltzmann law, radiation rate is proportional to $R^2T^4$. Substituting the given values gives a ratio of 4:1.

12. C) It relies on the principle that the electrical resistance of the metal changes uniformly with temperature. | Explanation: A platinum resistance thermometer works because the electrical resistance of platinum changes almost uniformly with temperature.

13. C) α = γ / 3 | Explanation: If the liquid level does not change, apparent expansion is zero. Hence γ = 3α, so α = γ/3.

14. B) 2 : 1 | Explanation: Specific heat is inversely proportional to the slope of the T–Q graph. Comparing the slopes gives the ratio 2:1.

15. C) It increases by a factor of √2. | Explanation: RMS speed depends only on absolute temperature. Doubling temperature increases RMS speed by √2.

16. A) $P_0V_0$ | Explanation: Net work done in a cycle equals the area enclosed by the P–V graph. The triangular area is $P_0V_0$.

17. B) β = (1 - η) / η | Explanation: Using Carnot relations, $\beta = T_2/(T_1-T_2)$. Replacing $T_2/T_1$ with (1−η) gives β = (1−η)/η.

18. D) It will remain unchanged | Explanation: Speed of sound in an ideal gas depends only on temperature. Since the process is isothermal, temperature stays constant, so the speed of sound remains unchanged.

19. B) 2:3 | Explanation: The first overtone of an open pipe is the 2nd harmonic (2f), while for a closed pipe it is the 3rd harmonic (3f). Therefore, the ratio is 2:3.

20. C) f | Explanation: Due to the Doppler effect, one frequency becomes 1.5f and the other becomes 0.5f. Their difference is f, which is the beat frequency.

21. A) f and 3I/4 | Explanation: Covering part of the lens reduces the amount of light but does not change its curvature. So focal length stays the same, while intensity becomes 3I/4.

22. B) 2.67 | Explanation: Fringe shift is proportional to $(\mu - 1)$. Replacing glass with water reduces the shift, giving about 2.67 fringe widths.

23. C) Increase the diameter of the objective lens | Explanation: Resolving power depends on the aperture diameter of the objective lens. A larger diameter gives better resolution.

24. A) $30^0$ | Explanation: At Brewster's angle, reflected and refracted rays are perpendicular. With $\mu = \sqrt{3}$, the refracted angle is 30°.

25. B) $f_1 = -2f_2$ | Explanation: For an achromatic combination, $\omega_1/f_1 + \omega_2/f_2 = 0$. Using $\omega_1 = 2\omega_2$ gives $f_1 = -2f_2$.

26. A) Two up quarks and one down quark (uud) | Explanation: A proton is made of two 'up' quarks (each with charge +2/3e) and one 'down' quark (charge -1/3e). The total charge is (+2/3) + (+2/3) + (-1/3) = +1e.

27. B) 1 - (1/e) | Explanation: The number of nuclei remaining after time t is given by $N = N_0 e^{-\frac{t}{T}}$. At $t = \tau$ (mean life), the remaining nuclei are $N = N_0 e^{-1} = N_0/e$. The question asks for the fraction that has DECAYED, which is $(N_0 - N)/N_0 = (N_0 - N_0/e)/N_0 = 1 - 1/e$.

28. B) More than double | Explanation: Einstein's photoelectric equation is $eV_s = h\nu - W$ (where W is the work function). Initially, $V_{s1} = (h\nu - W)/e$. If frequency is doubled, $V_{s2} = (2h\nu - W)/e = (2h\nu - 2W + W)/e = 2V_{s1} + W/e$. Because of the W/e term, the new stopping potential is more than double the original.

29. B) $2mc^2/e$ | Explanation: De-Broglie wavelength $\lambda_{deB} = h/\sqrt{(2meV)}$. The minimum X-ray wavelength (cutoff) $\lambda_{min} = hc/eV$. Equating the two: $h/\sqrt{(2meV)} = hc/eV$. Squaring both sides: $h^2/(2meV) = h^2c^2/(e^2V^2)$. Simplifying yields $1/(2m) = c^2/eV$, which rearranges to $V = 2mc^2/e$.

30. B) 4 | Explanation: It is a standard digital logic derivation that exactly 4 NAND gates are required to build a 2-input XOR gate. The configuration uses one NAND gate for the inputs A and B, whose output is fed into two separate NAND gates alongside A and B respectively, and their outputs feed into a final fourth NAND gate.

31. C) Iron-56 ($^{56}Fe$) | Explanation: The binding energy per nucleon peaks at around mass number A = 56 to 62. Iron-56 (along with Nickel-62) has the highest binding energy per nucleon (approx 8.8 MeV), making it the most thermodynamically stable nucleus in nature.

32. A) 5/27 | Explanation: Longest wavelength corresponds to the minimum energy transition. For Lyman, this is n=2 to n=1: $1/\lambda_L = R(1/1^2 - 1/2^2) = 3R/4 \Rightarrow \lambda_L = 4/3R$. For Balmer, it's n=3 to n=2: $1/\lambda_B = R(1/2^2 - 1/3^2) = 5R/36 \Rightarrow \lambda_B = 36/5R$. Ratio $\lambda_L/\lambda_B = (4/3R) / (36/5R) = 20/108 = 5/27$.

33. C) Kinetic Energy | Explanation: The magnetic force is always perpendicular to the velocity vector ($F = qv \times B$), meaning it acts as a centripetal force and does no work. Therefore, the speed and Kinetic Energy remain constant. Velocity, momentum, and acceleration are vector quantities and change constantly due to changing direction.

34. C) It increases. | Explanation: In reverse bias, the positive terminal of the battery is connected to the n-type and the negative to the p-type. This external field pulls the majority charge carriers (electrons in n, holes in p) away from the junction, thereby widening (increasing) the thickness of the depletion layer.

35. A) 1/9λ | Explanation: Using the decay law $N = N_0 e^{-\lambda t}$. For A, $N_A = N_0 e^{-10\lambda t}$. For B, $N_B = N_0 e^{-\lambda t}$. The ratio is $N_A/N_B = \frac{e^{-10\lambda t}}{e^{-\lambda t}} = e^{-9\lambda t}$.We are given $N_A/N_B = 1/e = e^{-1}$. Equating exponents: $-9\lambda t = -1$, which gives $t = 1/9\lambda$.

36. B) Directly proportional to its distance from Earth | Explanation: Hubble's Law is expressed as $v = H_0 \times d$, where v is the recessional velocity, d is the proper distance to the galaxy, and $H_0$ is Hubble's constant. This means the further away a galaxy is, the faster it is moving away from us (directly proportional).

37. C) It has gained exactly 3 electrons. | Explanation: Charge is quantized: Q = ne where $e = 1.6\times10^{-19}$ C. Thus, $n = (4.8\times10^{-19}) / (1.6\times10^{-19}) = 3$. Because the net charge is negative, the oil drop must have an excess of electrons. Therefore, it has gained 3 electrons.

38. B) F/3 (repulsive) | Explanation: Initially, the force is $F = k(q)(3q)/d^2 = 3kq^2/d^2$ (attractive). When they touch, the net charge is -2q, which divides equally so each sphere has -q. The new force is $F' = k(q)(q)/d^2 = kq^2/d^2$. Comparing the two, $F' = F/3$. Since both charges are now negative, the force is repulsive.

39. C) V is constant inside the sphere and decreases as 1/r otside. | Explanation: Inside a hollow conducting sphere, the electric field E = 0. Since E = -dV/dr, the potential V must be a constant everywhere inside, equal to its value on the surface. Outside the sphere, it behaves as a point charge, so $V \propto 1/r$.

40. C) $2\mu_0$ $and$ $2:1$ | Explanation: Since the battery remains connected, voltage V is constant. The initial energy density is $u_0 = \frac{1}{2} \varepsilon_0 E^2$. Because V and d are constant, E = V/d is constant. The new energy density $u = \frac{1}{2} (K\varepsilon_0) E^2 = Ku_0 = 2u_0$. Capacitance changes to C' = KC. Since Q = CV, the new charge is Q' = K(CV) = 2Q, making the ratio 2:1.

41. B) $6.2\mu F$ | Explanation: Capacitors C1 (2 μF) and C2 (3 μF) are in series. Their equivalent is $C_{12} = (2 \times 3) / (2 + 3) = 1.2$ μF. This combination is in parallel with C3 (5 μF). The total equivalent capacitance is $C_{total} = 1.2 + 5 = 6.2$ μF.

42. A) Only the 25W bulb will fuse | Explanation: Resistance of a bulb is $R = V^2/P$. The 25W bulb has higher resistance (1936 Ω) than the 100W bulb (484 Ω). When connected in series across 440V, the voltage divides in proportion to resistance. The voltage across the 25W bulb will be 440 × (1936 / (1936 + 484)) = 352V. Since 352V exceeds its rated 220V, the 25W bulb will fuse.

43. A) 2500Ω | Explanation: For the galvanometer to show zero deflection, the potential difference across the 500 Ω resistor must be exactly equal to the opposing battery's voltage (2V). Using the voltage divider rule for the main loop: $V_{500} = 12 \times (500 / (R + 500)) = 2V$. Solving this yields R = 2500 Ω.

44. B) $520^0$C | Explanation: The neutral temperature ($T_n$) is the arithmetic mean of the cold junction temperature ($T_c$) and the temperature of inversion ($T_i$). $T_n = (T_c + T_i) / 2 \Rightarrow 270 = (20 + T_i) / 2 \Rightarrow 540 = 20 + T_i \Rightarrow T_i = 520°C$.

45. B) 10 | Explanation: First, find the resonant angular frequency: $\omega_0 = 1 / \sqrt{LC} = 1 / \sqrt{0.2 \times 20 \times 10^{-6}} = 500$ rad/s. The Q-factor is given by $Q = (\omega_0 L) / R$. Substituting the values: Q = (500 × 0.2) / 10 = 100 / 10 = 10.

46. C) $627^0$C | Explanation: According to Curie's Law, magnetic susceptibility is inversely proportional to absolute temperature ($\chi \propto 1/T$). Initial temperature $T_1 = 27°C = 300K$. To reduce x to a third of its value, the absolute temperature must triple. $T_2 = 3 \times 300K = 900K$. Converting back to Celsius: 900 - 273 = 627°C.

47. B) The drift velocity of the charge carriers | Explanation: The Hall voltage $V_H$ is generated to balance the magnetic Lorentz force. At equilibrium, $qE = q v_d B$, which gives $V_H / d = v_d B$. Therefore, $V_H = v_d B d$, meaning it is directly proportional to the drift velocity ($v_d$) of the charge carriers.

48. B)  | Explanation: The radius of a circular path in a magnetic field is r = mv / (qB). Since Kinetic Energy $K = p^2 / (2m)$, momentum $p = \sqrt{2mK}$. Substituting this into the radius formula gives $r = \sqrt{2mK} / (qB)$. Because K, q, and B are the same for both the electron and proton, $r \propto \sqrt{m}$. Thus, the ratio is $\sqrt{m_e} : \sqrt{m_p}$.

49. A) $1/2\ B\omega L^2$ | Explanation: The induced motional emf across a rotating rod is $\frac{1}{2} B\omega L^2$. Applying the right-hand rule for the Lorentz force (F = q(v × B)), with B into the page and v tangential to rotation, positive charges are pushed towards the pivoted end (center), while free electrons are pushed to the outer free end. Therefore, the pivoted end becomes positive.

50. C) Laminated with thin sheets of steel insulated from each other | Explanation: Eddy currents are circulating currents induced within bulk pieces of conducting material. To minimize the power loss ($I^2R$) due to these currents, transformer cores are laminated with thin, varnished (insulated) sheets of steel. This breaks the path of the eddy currents, greatly increasing the effective resistance and reducing energy loss.

## CHEMISTRY

51. C) $C_4H_8O_2$ | Explanation: Mass of C = (12/44) × 0.88g = 0.24g. Mass of H = (2/18) × 0.36g = 0.04g. Mass of O = 0.44 - (0.24 + 0.04) = 0.16g. Moles ratio (C : H : O) = (0.24/12) : (0.04/1) : (0.16/16) = 0.02 : 0.04 : 0.01 = 2 : 4 : 1. The empirical formula is $C_2H_4O$ (Empirical mass = 44). Molecular mass = 2 × Vapor Density = 2 × 44 = 88. Ratio (n) = 88 / 44 = 2. Molecular formula = $(C_2H_4O)_2 = C_4H_8O_2$

52. B) 80% | Explanation: Reaction: $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$ . Moles of $CaCO_3$ = 20 / 100 = 0.2 mol. Moles of HCl = 1.0 M × 0.2 L = 0.2 mol. Since 1 mol $CaCO_3$ requires 2 mol HCl, HCl is the limiting reactant. 0.2 mol HCl will produce 0.1 mol $CO_2$ , Theoretical yield of $CO_2$ = 0.1 mol × 22.4 L/mol = 2.24 L. Percentage yield = (1.792 / 2.24) × 100% = 80%.

53. C) n = 3, l = 1, m = -1, s = +1/2 | Explanation: The energy of an electron in a multi-electron atom is determined by the n + l rule. Lower (n + l) means lower energy (more tightly bound). A) 3d: n+l = 5; B) 4s: n+l = 4; C) 3p: n+l = 4; D) 4p: n+l = 5. Between 4s and 3p, 3p has a lower principal quantum number (n=3), meaning it is deeper within the electron cloud and requires the highest energy to remove.

54. B) Radii decrease from $N^{3-}$ to $Al^{3+}$ due to an increasing effective nuclear charge pulling the same number of electrons closer. | Explanation: All species are isoelectronic (10 electrons). As we move from N (Z=7) to Al (Z=13), the number of protons increases. This increases the effective nuclear charge (Zeff). A stronger positive nucleus pulls the same 10 electrons closer, steadily decreasing the ionic radius.

55. B) Trigonal planar ($sp^2$), Trigonal pyramidal ($sp^3$), T-shaped ($sp^3d$) | Explanation:- $BF^3$ : 3 bond pairs, 0 lone pairs. Steric number = 3 → $sp^2$, Trigonal planar.- $NH_3$ : 3 bond pairs, 1 lone pair. Steric number = 4 → $sp^3$, Trigonal pyramidal.- $ClF^3$ : 3 bond pairs, 2 lone pairs. Steric number = 5 → $sp^3d$, T-shap

56. A) 392 | Explanation: In Mohr's salt, the active reducing species is $Fe^{2+}$. During the reaction with $KMnO_4$ , $Fe^{2+}$ oxidizes to $Fe^{3+}$. The change in oxidation state (n-factor) is |3 - 2| = 1. Equivalent weight = Molar mass / n-factor = 392 / 1 = 392.

57. B) Gas A has stronger intermolecular forces, deviates more from ideality, and is easier to liquefy. | Explanation: The critical temperature (Tc ) indicates the strength of intermolecular forces. A higher Tc (Gas A, 304 K) means the gas molecules attract each other strongly, causing greater deviation from ideal behavior and making the gas much easier to liquefy.

58. A) $AB_3C_2$ | Explanation:- Atoms at corners (A) = 8 × (1/8) = 1 atom.- Atoms at face centers (B) = 6 × (1/2) = 3 atoms.- Total atoms in the FCC lattice = 4. Number of tetrahedral voids = 2N = 8.- Atoms in tetrahedral voids (C) = 1/4 × 8 = 2 atoms. Empirical formula: $AB_3C_2$ .

59. B) The equilibrium will shift towards the reactants (backward reaction). | Explanation: Adding an inert gas at constant pressure increases the volume of the container (to maintain total pressure), decreasing the partial pressures of reacting gases. According to Le Chatelier's principle, the system shifts in the direction that produces more moles of gas. Since reactants have 4 moles and products have 2 moles, it shifts backward.

60. B) 50 mL | Explanation: Step 1: Find normality of the acid ($N_1V_1 = N_2V_2$). $N_{acid}$ × 50 = 0.1 × 40 → $N_{acid}$ = 0.08 N. Step 2: Neutralize $Na_2CO_3$ . 0.08 × $V_{acid}$ = 0.2 × 20 → 0.08 × $V_{acid}$ = 4 → $V_{acid}$ = 50 mL.

61. B) 4.74 | Explanation: Moles of $CH_3COOH$ = 100 mL × 0.1 M = 10 mol. Moles of NaOH = 50 L × 0.1 M = 5 mol. NaOH neutralizes half the acid, producing 5 mol of salt and leaving 5 mol of acid. This forms a buffer where [Salt] = [Acid]. pH = pKa + log([Salt]/[Acid]) = pKa + log(1) = pKa . pH = -log(1.8 × 10-5) = 5 - log(1.8) = 5 - 0.26 = 4.74.

62. B) $1.0 \times 10^{-9}$ M | Explanation: $BaSO_4(s) \rightleftharpoons Ba^{2+}(aq) + SO_4^{2-}(aq)$. Let solubility be s. $Na_2SO_4$ completely dissociates to provide 0.1 M of $SO_4^{2-}$. Total $[SO_4^{2-}]$ = 0.1 + s ≈ 0.1 M. $[Ba^{2+}]$ = s. Ksp = $[Ba^{2+}][SO_4^{2-}]$ → 1.0 × 10-10 = (s)(0.1) → s = 1.0 × 10-9 M.

63. C) Second order | Explanation: Half-life ($t_{1/2}$) relates to initial concentration (a) as $t_{1/2} \propto 1 / a_n$-1( $t_1$ / $t_2$ ) = ($a_2$ / $a_1$)(n-1) → 50 / 25 = (0.2 / 0.1)(n-1) → 2 = 2n-1 → n - 1 = 1 → n = 2.

64. C) 55 times | Explanation: The ratio of catalyzed rate to uncatalyzed rate is $k_{cat}$ / $k_{uncat}$ = e$\Delta E_a$ / RT. $\Delta E_a$ = 10,000 J/mol. T = 300 K. Exponent = 10000 / (8.314 × 300) = 10000 / 2494.2 ≈ 4.01. Rate increase factor = e4.01 ≈ e4 ≈ 54.6 ≈ 55 times.

65. B) 0.401 V | Explanation: Reaction: $Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$. n = 2. E°cell = E°cathode - E°anode = 0.80 - 0.34 = 0.46 V. $E_{cell}$ = $E°_{cell}$ - (0.0591 / 2) × log($[Cu^{2+}]$ / $[Ag^+]_2$) Ecell = 0.46 - 0.02955 × log(1.0 / 0.12) = 0.46 - 0.02955 × log(100). Ecell = 0.46 - (0.02955 × 2) = 0.46 - 0.0591 = 0.401 V.

66. A) -74.8 kJ/mol | Explanation: Formation reaction: C(graphite) + $2H_2(g) \rightarrow CH_4(g)$. ΔHf ($CH_4$) = ΣΔHc (Reactants) - ΣΔHc (Products) ΔHf ($CH_4$) = [ΔHc (C) + 2×ΔHc ($H_2$)] - [ΔHc ($CH_4$)] = [-393.5 + 2(-285.8)] - [-890.3] = -965.1 + 890.3 = -74.8 kJ/mol.

67. C) 11460 years | Explanation: Ratio of final activity (A) to initial activity (A0) determines the number of half-lives (n). A / $A_0$ = (1/2)n → 4.0 / 16.0 = 1/4 = (1/2)2. Two half-lives have passed. Total age = n × $t_{1/2}$ = 2 × 5730 = 11,460 years.

68. C) Free radical | Explanation: Homolytic bond fission involves the equal sharing of bonding electrons, resulting in species with an unpaired electron, known as free radicals.

69. C) sp | Explanation: The sp hybridized carbon in terminal alkynes has 50% s-character, making it highly electronegative and capable of releasing $H^+$ ions.

70. D) $(CH_3)_3CCl$ | Explanation: $SN^1$ reactivity depends on carbocation stability. $(CH_3)_3CCl$ forms a 3° carbocation, which is the most stable.

71. B) Ethanol | Explanation: Fermentation of sugar by yeast enzymes (invertase and zymase) yields ethanol and carbon dioxide.

72. B) Acetaldehyde | Explanation: Grignard reagents react with formaldehyde to give 1° alcohols, other aldehydes (like acetaldehyde) give 2° alcohols, and ketones give 3° alcohols.

73. B) Tert-butyl halide undergoes elimination | Explanation: Using a 3° alkyl halide with a strong base (alkoxide) leads to elimination (alkene formation) instead of $SN^2$ substitution due to severe steric hindrance.

74. B) Diethyl oxalate | Explanation: Hoffmann's method uses diethyl oxalate, which forms a solid with 1° amines, a liquid with 2° amines, and does not react with 3° amines.

75. B) Ketones | Explanation: Unlike Grignard reagents, the less reactive organocopper compounds stop at the ketone stage and do not further nucleophilically attack the resulting ketone.
76. B) Chloride > Anhydride > Ester > Amide | Explanation: Reactivity relies on the leaving group ability: Cl- > RCOO- > RO- > NH2-. Thus, acid chlorides are the most reactive.

77. D) Resonance stabilization of phenoxide | Explanation: The phenoxide ion formed after losing a proton is highly stabilized by the delocalization of the negative charge around the aromatic ring.

78. A) Propan-1-ol | Explanation: This reaction results in the anti-Markovnikov addition of water across the double bond without carbocation rearrangement, yielding a primary alcohol.

79. B) Iso-octane | Explanation: Iso-octane (2,2,4-trimethylpentane) burns smoothly and is arbitrarily assigned an octane number of 100, while n-heptane is assigned 0.

80. B) Partial double bond character of the C-Cl bond | Explanation: The lone pair on chlorine delocalizes into the benzene ring via resonance, giving the C-Cl bond partial double bond character, making it harder to break.

81. B) N-Phenylhydroxylamine | Explanation: Neutral reduction stops at the N-Phenylhydroxylamine stage. Acidic reduction would go all the way to aniline.

82. B) 1-Bromo-2-methylpropane | Explanation: The peroxide effect (Kharasch effect) causes HBr to add via a free-radical mechanism, resulting in anti-Markovnikov regioselectivity.

83. B) Lacks an alpha-hydrogen | Explanation: Aldehydes with no alpha-hydrogens cannot undergo aldol condensation; instead, they undergo self-oxidation-reduction (Cannizzaro) in concentrated base.

84. C) A Carboxylic acid | Explanation: The nucleophilic alkyl group of RMgX attacks the electrophilic carbon of $CO_2$, forming a carboxylate salt that yields a carboxylic acid upon protonation.

85. B) HF and HI | Explanation: HF has the highest boiling point due to strong intermolecular hydrogen bonding. HI is the strongest reducing agent because the H-I bond is the weakest, allowing it to easily release hydrogen.

86. C) It decomposes to form a basic flux that removes acidic gangue as slag. | Explanation: Limestone ($CaCO_3$) decomposes to CaO (a basic flux). The CaO then reacts with acidic gangue (like $SiO_2$) to form molten slag ($CaSiO_3$), which can be easily removed.

87. D) NO (Nitric oxide) | Explanation: Nitric oxide (NO), often released by supersonic jets, reacts catalytically with ozone ($O_3$) to form $NO_2$ and $O_2$, significantly contributing to ozone layer depletion.

88. A) -0.6 Δo | Explanation: In a high-spin $d_4$ complex, the electronic configuration is $t_2g(3)$ eg(1). The CFSE is calculated as [3 * (-0.4) + 1 * (0.6)]Δo = -1.2 + 0.6 = -0.6 Δo.

89. C) Cadmium (Cd) | Explanation: Itai-Itai disease is characterized by severe bone pain and osteomalacia, and it is primarily caused by long-term cadmium poisoning.

90. B) It acts as an oxidizing agent to convert Ag to $Ag^+$ so it can form a soluble complex. | Explanation: Oxygen oxidizes the elemental silver (or silver in $Ag_2S$) to Ag+, which then reacts with cyanide ions ($CN^-$) to form the soluble complex dicyanoargentate(I), $[Ag(CN)_2]^-$

91. C) A white precipitate forms which eventually turns grey. | Explanation: $HgCl_2$ is first reduced to calomel ($Hg_2Cl_2$), which is a white precipitate. Excess $SnCl_2$ further reduces the calomel to finely divided metallic mercury (Hg), which appears grey/ black.

92. B) Copper and Tin | Explanation: Poling involves stirring the molten metal with green wood logs. The hydrocarbons from the wood reduce the metal oxides (like $Cu_2O$ and $SnO_2$) back to the pure metals (Copper and Tin).

93. A) Formation of $Cr_2(SO_4)_3$, while $H_2S$ acts as a reducing agent. | Explanation: $H_2S$ acts as a reducing agent, reducing the orange dichromate ion ($Cr^{6+}$) to the green chromium(III) ion ($Cr^{3+}$). $H_2S$ itself is oxidized to elemental sulphur.

94. B) 3 $Na^+$ are pumped out of the cell, 2 $K^+$ are pumped in. | Explanation: The $Na^+/K^+$ ATP pump uses the energy from the hydrolysis of one ATP molecule to export three sodium ions (Na+) out of the cell and import two potassium ions ($K^+$) into the cell, maintaining the electrochemical gradient.

95. B) Vanadium(V) oxide ($V_2O_5$) | Explanation: Vanadium pentoxide ($V_2O_5$) is the preferred catalyst in the Contact process. Although platinum is technically more efficient, $V_2O_5$ is widely used in modern industry because it is significantly more cost-effective and less susceptible to catalytic poisoning by impurities such as arsenic compounds.

96. C) Nylon-6, 6 | Explanation: Nylon-6,6 is a condensation polymer formed by the reaction of hexamethylenediamine and adipic acid, with the elimination of water molecules during the polymerization process. Teflon, PVC, and polystyrene are all addition polymers formed directly from alkene derivatives without the loss of small molecules

97. A) NaOH/ $CO_2$ followed by $H^+$ , then $CH_3COCl$ | Explanation: This synthesis involves two major steps. First, Kolbe's reaction (treating phenol with NaOH and $CO_2$ , followed by acidification) converts phenol into salicylic acid (o-hydroxybenzoic acid). Second, acetylation of the phenolic hydroxyl group using acetyl chloride ($CH_3COCl$) or acetic anhydride yields acetylsalicylic acid, commonly known as aspirin.

98. C) Vacuum distillation | Explanation: Vacuum distillation (distillation under reduced pressure) lowers the ambient pressure above the liquid. Since a liquid boils when its vapor pressure equals the external pressure, reducing the external pressure allows the liquid to boil at a much lower temperature. This prevents thermal decomposition of unstable substances like glycerol or hydrogen peroxide during purification.

99. B) $Fe_4[Fe(CN)_6]_3$ | Explanation: During Lassaigne's test, nitrogen and carbon from the organic compound react with sodium metal to form sodium cyanide (NaCN). This reacts with added $FeSO_4$ to form sodium hexacyanoferrate(II). Upon addition of $Fe^{3+}$ ions (from $FeCl_3$), Iron(III) hexacyanoferrate(II) is formed, which has the formula Fe4 $[Fe(CN)_6]_3$ . This specific coordination complex is responsible for the Prussian blue color.

100. A) M | Explanation: In iodometric titration, the reaction is: $I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$. The oxidation state of sulfur changes from +2 in the thiosulfate ion to +2.5 in the tetrathionate ion. The total change in oxidation state for the two sulfur atoms in a single thiosulfate ion is 2 × (2.5 - 2) = 1. Since the n-factor (change in oxidation number per molecule) is exactly 1, the equivalent weight is equal to its molar mass (M / 1 = M)

## BOTANY

101) C) It increases Km but leaves Vmax unchanged. | Explanation: A competitive inhibitor competes with the substrate for the active site. It increases the apparent Km (more substrate is needed to reach half-maximal velocity) but does not alter the Vmax
because high concentrations of substrate can outcompete the inhibitor.

102) B) β-1,4-glycosidic linkages between glucose units | Explanation: Cellulose is a linear polymer of β-D-glucose units linked by β-1,4-glycosidic bonds. This β-linkage allows for flat, ribbon-like structures that pack tightly into microfibrils, making it structurally tough and resistant to enzymes like amylase (which break α-linkages). Biodiversity

103) B) Absence of a defined nucleus and membrane-bound organelles | Explanation: Kingdom Monera comprises prokaryotic organisms (bacteria and cyanobacteria). The defining feature of prokaryotes is the absence of a true nucleus (no nuclear membrane) and lack of membrane-bound organelles like mitochondria and chloroplasts.

104) B) Ascomycetes | Explanation: Yarsagumba is an entomopathogenic fungus that parasitizes the larvae of ghost moths. Taxonomically, it belongs to the phylum Ascomycota (Ascomycetes), as it produces its sexual spores (ascospores) in sac-like structures called asci.

105) B) Hypertension (high blood pressure) | Explanation: Rauwolfia serpentina contains the alkaloid reserpine, which acts as a sympatholytic agent. It is widely recognized and historically used for the management of hypertension and as a tranquilizer for certain psychiatric disorders.

106) B) Tetradynamous stamens and a false septum (replum) | Explanation: Brassicaceae (mustard family) is characterized by tetradynamous stamens (6 stamens: 4 inner long and 2 outer short) and parietal placentation where the ovary becomes bilocular due to the formation of a false septum called a replum.

107) B) Independent, haploid gametophytic phase | Explanation: In Dryopteris (a fern), the prothallus is a small, green, heart-shaped, independent structure that represents the haploid gametophyte stage. It bears the sex organs (antheridia and archegonia).

108) C) Haploid (n), formed before fertilization | Explanation: In gymnosperms like Pinus, the endosperm represents the female
gametophyte tissue. It is formed before fertilization occurs and is therefore haploid (n). This is unlike angiosperms, where endosperm is typically triploid (3n) and forms after double fertilization.

109) C) Laminarin and mannitol | Explanation: Brown algae (Phaeophyceae) store food as complex carbohydrates, primarily in the form of laminarin or mannitol. Floridean starch is found in red algae, and true starch is found in green algae and higher plants.

110) C) Nitrogen fixation | Explanation: Heterocysts are specialized, pale, thick-walled cells found in some cyanobacteria. They
maintain an anaerobic environment necessary for the functioning of the enzyme nitrogenase, thus playing a critical role in nitrogen fixation.

111) B) RNA and a protein coat | Explanation: Plant viruses typically consist of a single-stranded RNA core surrounded by a protein coat (capsid). Examples include the Tobacco Mosaic Virus (TMV). DNA plant viruses exist but are much less common. Ecology & Vegetation

112) C) Nitrobacter | Explanation: Nitrification is a two-step process. Ammonia is first oxidized to nitrite by bacteria like Nitrosomonas and Nitrococcus. The nitrite is then further oxidized to nitrate by Nitrobacter.

113) C) Phytoplankton | Explanation: The pioneer community in a hydrosere consists of microscopic phytoplankton (like diatoms and green algae) and zooplankton, which are the first to colonize the barren water body.

114) C) Sulfur dioxide and Nitrogen oxides | Explanation: Acid rain is caused mainly by sulfur dioxide ($SO_2$) and nitrogen oxides ($NO_x$) released from fossil fuel combustion. These gases react with water, oxygen, and other chemicals in the atmosphere to form sulfuric and nitric acids.

115) C) One species is harmed while the other is unaffected (- / 0) | Explanation: Amensalism is an interaction where one species is inhibited or harmed, while the other species remains entirely unaffected (- / 0). An example is the secretion of penicillin by Penicillium fungi, which kills bacteria but provides no immediate benefit to the fungus. Cell Biology

116) B) Lysosome | Explanation: Lysosomes are membrane-bound sacs containing acid hydrolases (enzymes that break down proteins, nucleic acids, lipids, and carbohydrates). These enzymes require an acidic environment (pH ~5) to be active.

117) B) S phase | Explanation: During the Synthesis (S) phase, DNA replication occurs, causing the DNA content to double (e.g., from 2C to 4C). However, the sister chromatids remain attached at the centromere, so the actual chromosome number remains the same (2n).

118) C) Pachytene | Explanation: Crossing over, mediated by the recombinase enzyme complex, occurs during the Pachytene stage of prophase I. The bivalents appear as tetrads at this stage. Zygotene is characterized by pairing (synapsis), and Diplotene by chiasmata visibility.

119) B) A fluid bilayer of phospholipids with embedded and peripheral proteins | Explanation: The Fluid Mosaic Model states that the membrane is a fluid, quasi-fluid phospholipid bilayer in which integral and peripheral proteins are interspersed like a mosaic, allowing lateral movement of molecules within the membrane.

120) C) They contain circular DNA and 70S ribosomes, similar to prokaryotes. | Explanation: Both mitochondria and chloroplasts have their own circular, double-stranded DNA and
70S ribosomes (like bacteria), dividing by a process similar to binary fission. This heavily supports the
theory that they originated as free-living prokaryotes engulfed by ancestral eukaryotic cells.
Genetics

121) C) To carry specific amino acids to the ribosome during translation | Explanation: tRNA acts as an adapter molecule. It has an anticodon loop that pairs with the mRNA codon and an amino acid acceptor end that carries the specific amino acid corresponding to that codon to the ribosome during protein synthesis.

122) C) Incomplete linkage | Explanation: Linked genes tend to be inherited together because they are on the same chromosome.

In incomplete linkage, crossing over occurs occasionally, producing some recombinant types, but parental types appear at a much higher frequency than the expected independent assortment ratio.

123) C) 50% | Explanation: Color blindness is X-linked recessive. The woman's father was color-blind ($X^CY$), so she must be a carrier ($X^CX^C$). The man is color-blind ($X^CY$). Their sons inherit Y from the father and either $X^C$ or $X^C$ from the mother. Thus, there is a 50% chance a son will inherit $X^C$ and be color-blind.

124) B) Monosomy | Explanation: Aneuploidy is the gain or loss of chromosomes. Monosomy is the loss of a single chromosome (2n - 1). Turner's syndrome in humans (45, XO) is an example of monosomy. Nullisomy is 2n-2, Trisomy is 2n+1.

125) B) 47, XXY | Explanation: Klinefelter's syndrome occurs in males who have an extra X chromosome, resulting in a karyotype of 47, XXY. It leads to symptoms like underdeveloped testes, sterility, and sometimes gynecomastia (breast development).

126) B) 30% | Explanation: Chargaff's rule states that in double-stranded DNA, Adenine (A) pairs with Thymine (T), and Guanine (G) pairs with Cytosine (C). If A = 20%, then T = 20% (total 40%). The remaining 60% must be G and C. Since G = C, Cytosine is 60% / 2 = 30%.n Plant Anatomy

127) B) Collenchyma | Explanation: Collenchyma consists of living cells with walls thickened at the corners by cellulose, hemicellulose, and pectin. It provides flexibility and mechanical strength to young stems and leaf petioles. Sclerenchyma provides support but consists of dead, heavily lignified cells.

128) B) Scattered, collateral, and closed with a bundle sheath | Explanation: Monocot stems have vascular bundles scattered throughout the ground tissue. They are conjoint, collateral, and closed (lacking cambium, thus no secondary growth). They are typically surrounded by a sclerenchymatous bundle sheath, and the xylem often takes a 'Y' or 'V' shape.

129) C) Endodermis of the root | Explanation: Casparian strips are band-like thickenings composed of water-impermeable suberin
found on the radial and transverse walls of the endodermis in roots. They force water to enter the living protoplast (symplastic pathway) before reaching the xylem. Plant Physiology

130) B) DPD is 10 atm; water will enter the cell. | Explanation: DPD (Suction Pressure) = OP - TP. Here, DPD = 15 - 5 = 10 atm. Because pure water has a DPD of 0 atm, and water moves from an area of lower DPD to higher DPD, water will flow into the cell until TP equals OP (making DPD zero).

131) B) Phosphoenolpyruvate (PEP) | Explanation: In $C_4$ plants, the initial fixation of $CO_2$ occurs in the mesophyll cells where
Phosphoenolpyruvate (PEP), a 3-carbon compound, accepts $CO_2$ in the presence of PEP carboxylase to form the 4-carbon compound Oxaloacetic acid (OAA).

132) C) Potassium ($K^+$) | Explanation: The active potassium pump hypothesis states that stomatal opening is caused by the active influx of $K^+$ ions into guard cells, which lowers their water potential, causing water to enter via osmosis. This increases turgor pressure, opening the pore.

133) C) ATP, NADPH, and $O_2$ | Explanation: Non-cyclic photophosphorylation involves both Photosystem II and Photosystem I. It results in the photolysis of water (releasing $O_2$), the generation of ATP via an electron transport chain, and the reduction of $NADP^+$ to NADPH. Cyclic photophosphorylation only produces ATP.

134) C) Mitochondrial matrix | Explanation: The Krebs cycle occurs in the mitochondrial matrix. Glycolysis occurs in the cytoplasm, and the Electron Transport System (ETS) is located on the inner mitochondrial membrane (cristae).

135) B) Gibberellin | Explanation: Gibberellins (GAs) stimulate massive stem elongation in rosette plants (like cabbage and beet) just before they flower. This rapid internode elongation is termed 'bolting'. Developmental Botany

136) B) A diploid zygote and a triploid primary endosperm nucleus (PEN) | Explanation: Double fertilization involves syngamy (fusion of one male gamete with the egg to form a diploid 2n zygote) and triple fusion (fusion of the second male gamete with two polar nuclei to form a triploid 3n Primary Endosperm Nucleus).

137) A) 12 and 36 | Explanation: The leaf cell is a diploid somatic cell (2n = 24), so the haploid number (n) is 12. Synergids are part of the female gametophyte and are haploid (n = 12). The PEN in angiosperms is formed by triple fusion and is triploid (3n = 36). Applied Botany

138) B) Callus | Explanation: When a plant explant is cultured on a nutrient medium containing appropriate hormones (auxin and cytokinin), the cells divide to form an amorphous, unorganized, and undifferentiated mass called a callus.

139) B) Agrobacterium tumefaciens | Explanation: Agrobacterium tumefaciens is a soil bacterium that naturally infects dicot plants, transferring its T-DNA from the Ti (Tumor-inducing) plasmid into the plant's nuclear DNA. Scientistsmodify this plasmid to deliver desirable genes into plants.

140) B) Anabaena azollae | Explanation: Azolla leaves contain cavities that host the cyanobacterium Anabaena azollae. This symbiotic relationship fixes atmospheric nitrogen, greatly enriching the soil in flooded rice paddies without the need for chemical nitrogen fertilizers.

## ZOOLOGY

141) B) Oxygen | Explanation: The primitive atmosphere was reducing because free oxygen ($O_2$) was absent. All available oxygen was bound in compounds like water vapor.

142) B) Amino acids | Explanation: The Miller-Urey experiment successfully synthesized simple organic molecules, predominantly amino acids, from a mixture of inorganic gases.

143) B) Australopithecus | Explanation: While Ramapithecus was an early ape-like ancestor, Australopithecus is recognized as the first hominid to be fully bipedal (walking upright on two legs).

144) B) Type of locomotory organelles | Explanation: Protozoans are classified into groups like Sarcodina, Mastigophora, and Ciliata based on their locomotory structures (pseudopodia, flagella, cilia).

145) B) Presence of a notochord | Explanation: The fundamental diagnostic features of chordates include a notochord, a dorsal hollow nerve cord, and pharyngeal gill slits at some stage of life.

146) C) Mammary glands and hair | Explanation: The presence of milk-producing mammary glands and body hair are exclusive diagnostic features of the class Mammalia.

147) C) Mollusca | Explanation: A soft, unsegmented body, a muscular foot for locomotion, and a visceral mass covered by a mantle (and often a shell) are diagnostic of Mollusca.

148) C) Columnar epithelium | Explanation: Columnar epithelium consists of tall, pillar-like cells specifically adapted for secretion (e.g., mucus) and absorption, primarily found in the GI tract.

149) C) Dense connective tissue | Explanation: Tendons (muscle to bone) and ligaments (bone to bone) are made of dense, fibrous connective tissue for strength.

150) C) Heart | Explanation: Cardiac muscle is unique: it is structurally striated but functions involuntarily, and possesses intercalated discs for synchronized beating.

151) A) Neuron | Explanation: The neuron is the core functional unit of the nervous system, transmitting electrical signals. Neuroglia are supporting cells.

152) B) Sporozoite | Explanation: Sporozoites are the motile, infective form of Plasmodium stored in the mosquito's salivary glands and injected into the human host.

153) D) Plasmodium falciparum | Explanation: P. falciparum causes malignant tertian malaria, the most dangerous form due to complications like cerebral malaria.

154) B) Segments 14, 15, 16 | Explanation: In a mature earthworm, the clitellum covers segments 14-16 and secretes a cocoon for holding eggs.

155) B) Improves soil fertility and aeration through burrowing and casting | Explanation: Earthworms aerate the soil by burrowing and enrich it with their nitrogen-rich excreta (castings).

156) C) Cutaneous respiration (skin) | Explanation: During hibernation, the frog's metabolism slows down, and its oxygen needs are met entirely by gas exchange through its moist skin.

157) B) Two atria and one ventricle | Explanation: Amphibians like frogs have a three-chambered heart, leading to a mixing of oxygenated and deoxygenated blood in the single ventricle.

158) C) Fats | Explanation: The liver secretes bile, which is essential for the emulsification of fats, breaking them down so pancreatic lipase can digest them.

159) B) Pancreas and Liver | Explanation: Alkaline bile from the liver and bicarbonate-rich pancreatic juice neutralize the acidic chyme entering the duodenum.

160) C) Bicarbonate ions in plasma | Explanation: About 70% of $CO_2$ is transported as bicarbonate ions ($HCO_3$) in the blood plasma.

161) C) Emphysema | Explanation: Emphysema damages the alveolar sacs, reducing the surface area for gas exchange, primarily caused by smoking.

162) B) Pumped out by each ventricle per minute | Explanation: Cardiac output = Stroke Volume □ Heart Rate, measuring the volume of blood pumped per ventricle in one minute.

163) B) Lacks both Anti-A and Anti-B antibodies in the plasma | Explanation: Since AB individuals have both A and B antigens, their plasma lacks antibodies against them, preventing rejection of transfused blood.

164) C) Bowman's capsule and Glomerulus | Explanation: Blood is filtered under high pressure from the glomerulus capillaries into the Bowman's capsule, initiating urine formation.

165) B) Endocrine pancreas (Diabetes mellitus) | Explanation: Lack of insulin from the pancreas prevents cells from taking up glucose, causing it to spill over into the urine.

166) B) Thalamus | Explanation: The thalamus acts as the brain's main relay station, directing sensory and motor signals to the appropriate areas of the cerebral cortex.

167) C) Sodium ($Na^+$) | Explanation: A nerve impulse is triggered when voltage-gated sodium channels open, allowing $Na^+$ to rush in and depolarize the cell.

168) C) Iris | Explanation: The iris is the muscular ring that controls the size of the pupil, thus regulating how much light enters the eye.

169) A) Thyroid gland | Explanation: The thyroid gland secretes hormones like thyroxine (T4), which strictly control the body's basal metabolic rate.

170) C) Iodine | Explanation: Iodine is structurally required to synthesize thyroid hormones. A deficiency causes the gland to swell (goiter) in an attempt to capture more iodine.

171) B) Seminiferous tubules | Explanation: Spermatogenesis (sperm production) takes place within the highly coiled seminiferous tubules inside the testes.

172) C) Luteinizing Hormone (LH) | Explanation: The rapid peak (LH surge) triggers the rupture of the mature Graafian follicle, releasing the ovum.

173) B) TB (Tuberculosis) | Explanation: TB is caused by the bacterium Mycobacterium tuberculosis, typically attacking the lungs.

174) C) Fungus | Explanation: Candidiasis (like thrush) is caused by Candida albicans, a type of yeast/fungus.

175) C) Natural active acquired immunity | Explanation: It is "natural" because it was caused by a real infection, and "active" because the body's own immune system generated the antibodies.

176) B) Live attenuated vaccine | Explanation: The oral polio vaccine (Sabin) uses a weakened (attenuated) form of the live virus.

177) B) Detecting genetic and chromosomal abnormalities in a fetus | Explanation: Amniocentesis involves extracting amniotic fluid to analyze fetal cells for chromosomal disorders like Down syndrome.

178) C) Heterotrophic microbes (bacteria and fungi) | Explanation: In sewage treatment, aerobic and anaerobic heterotrophic bacteria are used to naturally consume and break down organic pollutants.

179) C) Taxis | Explanation: Taxis is an innate behavioral response involving movement directly toward or away from a stimulus (e.g., phototaxis).

180) D) Biodiversity Hotspots | Explanation: Biodiversity hotspots are conservation priority areas characterized by exceptional levels of plant endemism and massive habitat loss

## MAT

181. B) Eyes | Explanation : The relationship connects a specific clinical disorder directly to the major anatomical organ it structurally compromises. Pneumonia affects the Lungs. Symmetrically, Glaucoma is an ophthalmic pathology defined by elevated intraocular pressure that directly compromises the Eyes.

182. C) Congregate | Explanation : Disseminate, Disperse, and Circulate are semantic synonyms sharing an identical directional vector: moving specific elements outward. Conversely, Congregate represents the exact opposite vector, meaning to assemble or group separate individual entities inward.

183. A) JHEQZO | Explanation : The conversion rule adds an expanding shift value to each alphabet index: New_letter = Current_letter + (index + 1). 'CANDLE' maps as: +2, +3, +4, +5, +6, +7. Applying this exact progressive matrix to 'HEALTH': H(+2)=J, E(+3)=H, A(+4)=E, L(+5)=Q, T(+6)=Z, H(+7)=O

184. C) Son | Explanation : "Her mother's only biological daughter" simplifies directly to the woman herself. "My maternal grandmother's only daughter" simplifies exclusively to Rohit's biological mother. Equating both lines establishes that the woman in the picture is Rohit's mother, making Rohit her Son

185. B) 24 | Explanation :

186. B) 185 | Explanation: The sequence values grow based on an escalating arithmetic addition-multiplier rule: $X_{n+1} = (X_n \times 2) + n$. (90 × 2) + 5 = 185.

187. B) 7,200 samples | Explanation : The progression follows standard geometric compounding: Target = Base × $(1 + r)^t$. 5000 × 1.20 = 6000 for year 1. 6000 × 1.20 = 7,200 for year 2

188. C) 85 | Explanation : The algebraic equation governing every row is: $(\text{Column}_1)^2 - (\text{Column}_2)^2$ = Column_3. For Row 3: $11^2 - 6^2 = 121- 36 = 85$.

189. B) 6 days | Explanation : Apply the compound work rate formula: $(M_1 \times D_2) / W_1 = (M_2 \times D_2) / W_2$. (3 × 4) / 60 = (5 × $D_2$) / 150. Solving for $D_2$ yields 6 days.

190. B) 5,346 | Explanation : The numbers follow a progressive polynomial function rule: $X_{n+1} = (X_n \times n) + n^2$. Target Step: $(885 \times 6) + 6^2 = 5310 + 36 = 5{,}346$.

191. B) 3 Km East | Explanation : Track coordinates on a 2D grid: Moving 4 km North (0,4). Turn right to go East 3 km (3,4). Turn right to go South 4 km (3,0). The displacement from (0,0) to (3,0) is exactly 3 km East.

192. B) $24^{th}$ | Explanation : Total Count = Index_Left + Index_Right - 1. So, 35 = 12 + Index_Right - 1. Solving gives Index_Right = 24.

193. A) Executive B | Explanation : Evaluating the constraints places the executives in the clockwise sequence: D → E → B → A → C. Facing the center from position A, the person sitting to the immediate right is Executive B.

194. A) Sunday | Explanation : "Two days prior to Friday" translates to Wednesday. Thus, "the day following tomorrow" is Wednesday. This means tomorrow is Tuesday, and today is Monday. Therefore, yesterday was Sunday

195. B) Product Gamma in march | Explanation : Using the exclusion grid: Sita launches in February. Ramesh manages Alpha (not in March), so Ramesh launches Alpha in January. This leaves March for Hari. Since Sita didn't manage Gamma, Hari must manage Product Gamma.

196. C) Face 5 | Explanation : In a continuous unfolded 2D cross layout of a 3D cube, face pairings that are separated by exactly one intervening cell square alternate to form opposite face planes. Face 3 is separated by exactly one middle cell (Face 4) from Face 5.

197. A | Explanation : A vertical mirror plane flips all components horizontally across the axis. The internal blue arrow triangle pointing rightward reflects to point leftward. The red accent dot sits in the top-left quadrant (the furthest distance from the mirror); its reflection maps to the top-right quadrant as seen in option (A)

198. B) A single centered circular hole located exactly at the coordinate origin of the paper sheet. | Explanation: The double fold brings the center of the paper exactly to the top-right corner of the final folded square layer. Executing a hole cut through this specific corner slices directly through the unified core center of the unfolded paper sheet

199. B) Figure ‘B’ | Explanation : Option Figure (B) houses the target configuration embedded cleanly inside its lines. Superimposing the exact vector coordinates over the lines in Option Figure (B) verifies that it houses the top line, bottom line, cross line, and Z diagonal perfectly.

200. B | Explanation: Tracking the three variables: Outer dot shifts clockwise to the Bottom-Left corner. The internal vector rotates clockwise by 45 degrees to a Diagonal Down-Right orientation (backslash "). The central Plus Sign (+) alternates ON/OFF and must be completely absent. This perfectly matches option (B